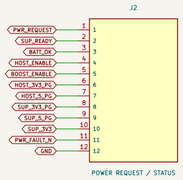
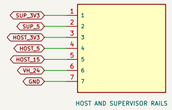
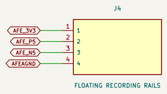
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

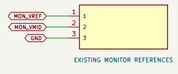
2S protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

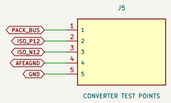












SUPPLIED
PACK_POS

SUPPLIED
PACK_BUS

SUPPLIED
GND

SUPPLIED
SUP_3V3

SUPPLIED
SUP_5

SUPPLIED
HOST_3V3

SUPPLIED
HOST_5

SUPPLIED
HOST_15

SUPPLIED
VM_24

input
File: input.kicad_sch

sup3
File: sup3.kicad_sch

sup5
File: sup5.kicad_sch

host3
File: host3.kicad_sch

host5
File: host5.kicad_sch

enable
File: enable.kicad_sch

enable_2
File: enable_2.kicad_sch

enable_3
File: enable_3.kicad_sch

boost15
File: boost15.kicad_sch

boost24
File: boost24.kicad_sch

island
File: island.kicad_sch

afe3
File: afe3.kicad_sch

afep
File: afep.kicad_sch

afen
File: afen.kicad_sch

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /
File: power.kicad_sch

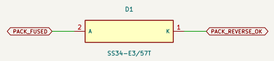
Title: Project SCCI RH S4 / Interfaces and rail handoff

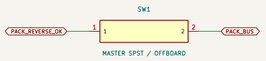
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

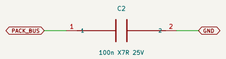
25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.











All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

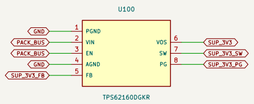
Sheet: /input/
File: input.kicad_sch

Title: Project SCCI RH S4 / Protected pack and input disconnect

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

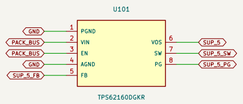
Sheet: /sup3/
File: sup3.kicad_sch

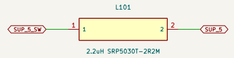
Title: Project SCCI RH S4 / Always-on supervisor 3.3 V

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

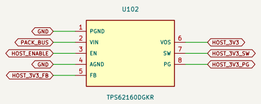
Sheet: /sup5/
File: sup5.kicad_sch

Title: Project SCCI RH S4 / Always-on supervisor 5 V

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

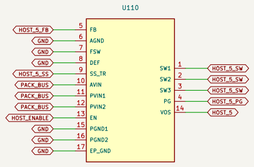
Sheet: /host3/
File: host3.kicad_sch

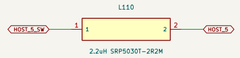
Title: Project SCCI RH S4 / Host logic 3.3 V

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.

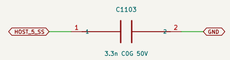




















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

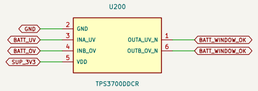
Sheet: /host5/
File: host5.kicad_sch

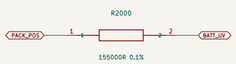
Title: Project SCCI RH S4 / Host and module input 5 V

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 6/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.





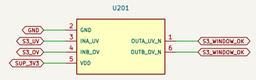










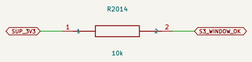


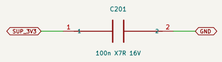


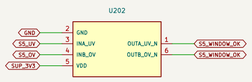








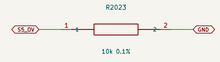














All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /enable/
File: enable.kicad_sch

Title: Project SCCI RH S4 / Native battery and startup qualification

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 7/15

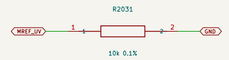
DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







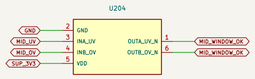












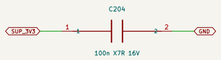






















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

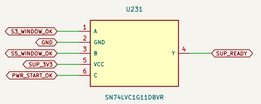
Sheet: /enable_2/
File: enable_2.kicad_sch

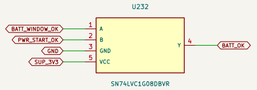
Title: Project SCCI RH S4 / Native battery and startup qualification / 2

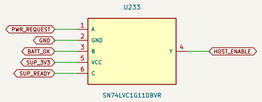
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 8/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.















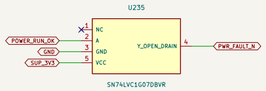




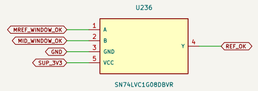


















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /enable_3/
File: enable_3.kicad_sch

Title: Project SCCI RH S4 / Native battery and startup qualification / 3

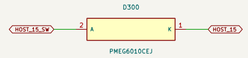
Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 9/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.



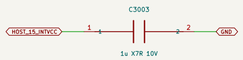




















All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /boost15/
File: boost15.kicad_sch

Title: Project SCCI RH S4 / Host 15 V boost

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 10/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.























All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /boost24/
File: boost24.kicad_sch

Title: Project SCCI RH S4 / Stimulation 24 V boost

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 11/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.









All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

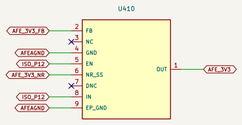
Sheet: /island/
File: island.kicad_sch

Title: Project SCCI RH S4 / Floating recording power module

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 12/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

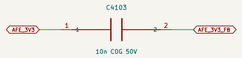
25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







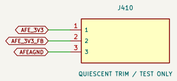












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

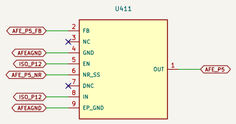
Sheet: /afe3/
File: afe3.kicad_sch

Title: Project SCCI RH S4 / Floating AFE 3.45 V regulator

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 13/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







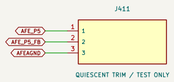












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power; firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

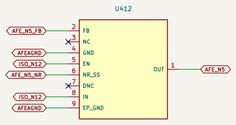
Sheet: /afep/
File: afep.kicad_sch

Title: Project SCCI RH S4 / Floating AFE +5.30 V regulator

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 14/15

DIGITAL SCCI ENGINEERING / ARTIFICIAL LOADS / BATTERY AND POWER

25 protected pack, independent supervisor rails, host converters and floating AFE supplies. Digital circuit design, not implant qualification.







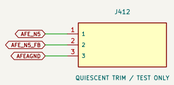












All labels are native electrical connections. PG is diagnostic; full qualification remains the nine independent supervision windows.

Converter shutdown does not discharge boost paths or remove stored energy. Body-facing shutdown remains the separate terminal guards.

Native power: firmware, PCB layout, EMI and physical battery/thermal qualification remain

Project SCCI RH

Sheet: /afen/
File: afen.kicad_sch

Title: Project SCCI RH S4 / floating AFE −5.10 V regulator

Size: A2	Date: 2026-10-14	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 15/15